

LEARN IMPACTIVE JACQUARD



**MACHINE KNIT
WITH KNIT-1**

Impactive Jacquards



Impactive Jacquards

This book includes:

- ❖ How to knit jacquard, graphic 2 colour jacquards.
- ❖ Manipulate design with card lock lever, colour changing.
- ❖ Mirroring, single motif, intarsia method, breaking stripes with pattern.
- ❖ Tuck stitch, slip stitch, mosaics, engineering the pattern.
- ❖ Manipulation of fabric, single colour textural jacquards.
- ❖ 14 Fully explained techniques.
- ❖ 26 design/swatch examples.

Provides references to relevant artists and fashion designers to further develop your creativity in jacquard design.

“I can’t wait to show everyone my new skills back at Uni”- Jess Northumbria UK.

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Jacquard is one of the most exciting kinds of machine knitting and easy to use to create strong patterns.

The design can be a picture, motif, or all over pattern or a combination of all three.

We are going to concentrate on all the possibilities two colour jacquard offers us.



The swatches on this page reference the traditional designs of Scandinavian

knitting usually either, red, blue, black with white.

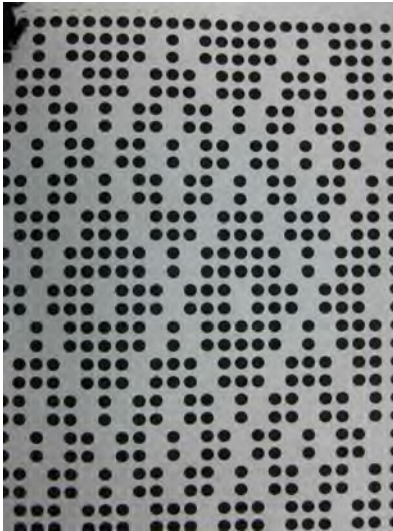
Geometric patterns of stars, snowflakes, reindeers, diamonds-used in all over patterns or as borders/yokes/cuffs.

If you like this aesthetic the historical knitting of Sweden, Norway, Denmark and the northernmost Scottish islands are worth exploring for inspiration.





Punchcards



- ❖ The best way to design a two colour jacquard is to draw it on graph paper, using one square for one stitch and one row. Fill in the squares of your pattern and leave the background blank. Then transfer to a punchcard and punch out your design.

Jacquard fabric is essentially a kind of slip stitch- the background yarn slipping behind those needles selected for the design and forming a loop or a 'float' and vice versa with the contrast yarn.

- ❖ On your punchcard, the punched holes on each horizontal row cause corresponding needles to be selected into patterning position and to knit; the unpunched spaces cause the remaining needles to stay in working position and act according to the cam setting(so jacquard, slip or tuck).

I'm using a 5 gauge Brother machine. If you have pre punched cards that came with your machine, use one of those for this first try until it becomes clear, then design your own.

How to Knit a Jacquard

- ❖ On your machine select 48 needles (24 either side of 0)

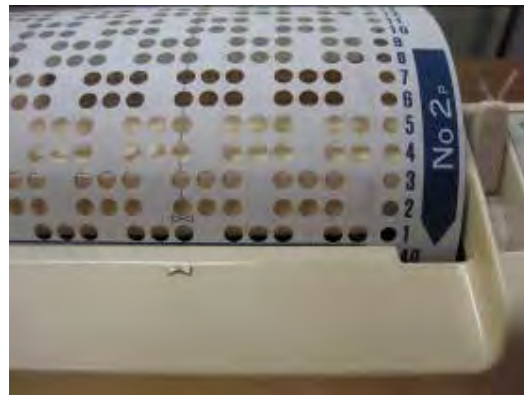
Have two yarns threaded through your tension feeder (antennae) Yarn A and a second colour Yarn B. A and B should be contrasting so it's easy for you to see the pattern clearly - try black and white to start.





❖ Cast on and knit some plain knitting in Yarn A your main colour.

❖ It might be helpful to make a photocopy of your punchcard so you can see what's happening as you knit it-You may need this later if you want to knit certain colours in certain places for example.



❖ Feed your card into the slot and turn the feeding knob (to the side of it) push down and turn clockwise until the starting line of the pattern. IT MUST SIT STRAIGHT.



The lever to the side of your punchcard controls how you knit it. It's the card lock lever.

❖ At this stage make sure the lever is on the centre triangle position, exactly like the photo right.

❖ With your carriage on the right turn the change knob to KC



❖ Knit one row to the left. You can feel and hear the needles being selected. Now you should have something like the photo left (still just 1 colour).





On Brother machines the needle selection from the punchcard is made directly on the needles so the selection for the next row is visible, (you can see where you are on your pattern by corresponding the card or photocopy of your card to the needles that have been selected) You can also change the selection manually before the row is knitted.



- ❖ Take your Yarn B and thread it into position B on your carriage as in the photo.
- ❖ Push in the MC cam.
- ❖ Knit back to the right and you should have something like this below.
- ❖ And that's it!

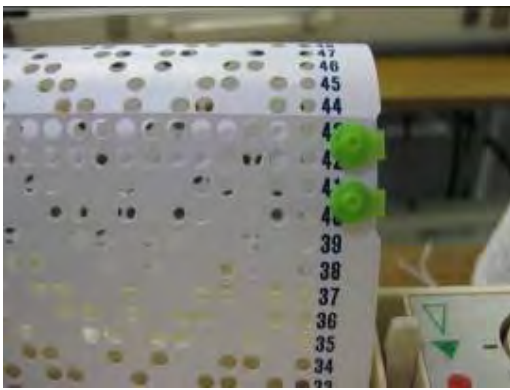
Keep knitting like this until you have knitted all your pattern, you will know when there is no more pattern when the carriage selects all the needles.



- ❖ Turn the change knob from KC to N.L or N. This will automatically push out the MC cam. Take the second yarn out from pos. B and knit plain knit Yarn A and take a look at your pattern.
- ❖ On a Brother machine the pattern on the 24 st. punchcard will be knitted on the centre section of the needle bed- 12 needles on either side of zero- and on consecutive 24- needle sections to each end of the bed. The position is fixed and cannot be moved.

You will now notice how the design repeats horizontally. Be very aware of this when designing your punchcard, a photocopier is very useful in this instance too.





If you want to knit your punchcard continuously, to repeat vertically, these handy little plastic clips should be with your machine in the accessories compartment.

Join the beginning and the end of the card together with these at either edge and keep knitting, they will go through the machine's mechanism, no problem.

The Card Lock Lever



When the lever is at this point – the circle – the card is locked.

You can knit but the card will not move- so your pattern will look something like this.



The needle selection will not change.

If you want to knit plain knit, one colour, in between patterns you;

- ❖ Set the card lock lever to lock, knit plain.
- ❖ Set the lever back to the coloured triangle to continue using punchcard.



When the lever is at this position- the empty triangle-the punchcard will only move when carriage is moved from right to left.



- ❖ So you will effectively be knitting two rows on every row on your punchcard so the design becomes double its original length.

- ❖ On standard machines it is only possible to mirror on the horizontal axis so; punch one vertical repeat of the design on the card.
- ❖ To mirror it, remove the card after the first rotation is complete, turn the card backward and upside down and knit the next rotation.
- ❖ Depending on where you stop and turn the card, and depending on the pattern you may be able to line it up and create new shapes as in the photo.





This is the same card mirrored in a different place and a stripe of different yarns a tone lighter.



- ❖ Changing colour cleverly as you knit a jacquard is the easiest way to make it more impactive.

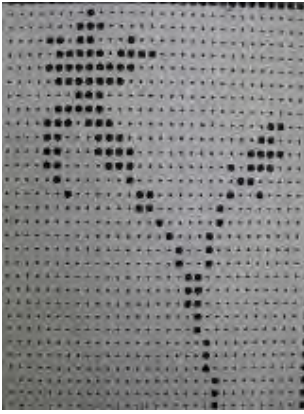
- ❖ Combine all the techniques you've encountered into a design.
- ❖ Remember your pattern doesn't have to be all over your fabric, combine with some plain knit.
- ❖ You might want to take a look at fashion designers like Anna Sui or Dries Van Noten for inspiration for pattern and use of jacquard knit.



- ❖ For geometric or tessellated patterns, take inspiration from the artist M.C.Escher
- ❖ A horizontal one-way design such as letters of the alphabet must be punched in reverse in order to appear correct in the finished knitting.



Single Motif



If you want to knit a single motif, for example the design to the left, use a slightly different method.

To knit this as a jacquard would give you enormous floats on the back of your fabric, so you only bring in yarn B exactly where you want your single motif.

- ❖ Set up the punchcard as usual.
- ❖ KC and your needles will be selected.
- ❖ So if you look at the pattern above, it starts with only one needle for the stem.
- ❖ 2 needles have been selected as the pattern wants to repeat. The very end needles are always selected to catch Yarn B



- ❖ Yarn A and Yarn B are in place MC pushed in.



- ❖ Now I don't want any floats, so I simply push back to B position all except the one needle for my single motif.
- ❖ So push back the end needles and any repeat.
- ❖ Knit the row and repeat- you will have to push the needles back *each* row manually.



As there are not needles being selected across the bed it's relatively easy to see where you use yarn B or change colour yarns.

Intarsia method

This intarsia method also means you can change the colour of one or more of your motifs so you can knit 3 or more colours in a row. As sample to the right- I have knitted 2 flowers in grey, one in black.



This is the wrong side of the swatch. You can see there are no floats between the grey and the black flowers.

- ❖ So in each row you knit the first motif, take yarn B out of feeder B on the carriage. Move the carriage across slowly!



- ❖ See the next needle selection.
- ❖ Change your yarn to the next colour and put in feeder B on the carriage.
- ❖ As you are knitting the motifs separately, there are no floats in between.
- ❖ Repeat slowly to finish.



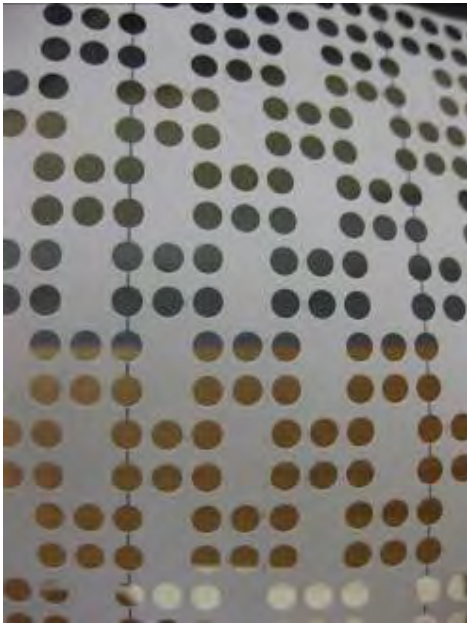


This design uses a single motif jacquard!



This design uses a repeated motif jacquard!

Variation on one Punchcard



So with one punchcard you can get many different patterns.

- ❖ Take this example a pre punched card that came with the machine.
- ❖ So the white yarn is colour A and the black colour B.



- ❖ Here I have knitted it as a jacquard and then elongated it, using the card lock lever on lock then the empty triangle position.



- ❖ The yarn is lambswool so I machine washed the swatch so it felts together. It now looks like this on the right side.....

-and on the wrong side I have cut the floats with scissors (it will be quite stable because it's been washed) so it looks like this.





Still using the same punchcard, I have used the design as blocks by;
(From the bottom)



- ❖ Yarn A white Yarn B black card is locked.
- ❖ Change knob to KC and push in MC cam, knit 4 rows.
- ❖ Take off KC and knit plain black.
- ❖ Now repeat but Yarn A is black, Yarn B is white.



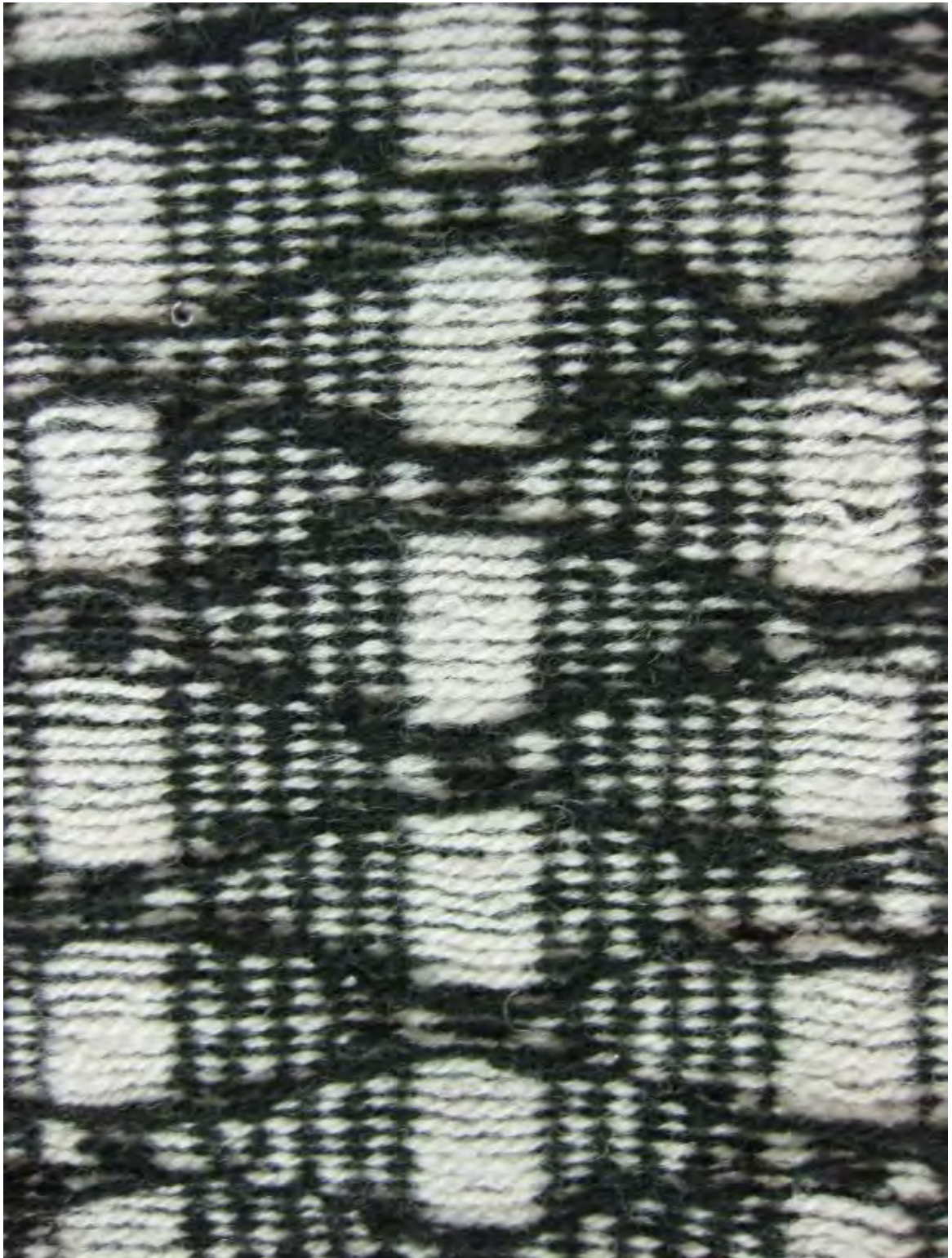
- ❖ Keep repeating this process of swapping A and B yarns and knitting plain stripes in between.

This is a variation with more jacquard rows.



Another example of using the jacquard design as a stripe.

Chanel and James Miller have used this technique very effectively in recent collections.



❖ If we start to think how to use these patterned pieces on the body ...

I knitted a long jacquard sample then hooked the selvedge (side edge) back on to the machine.



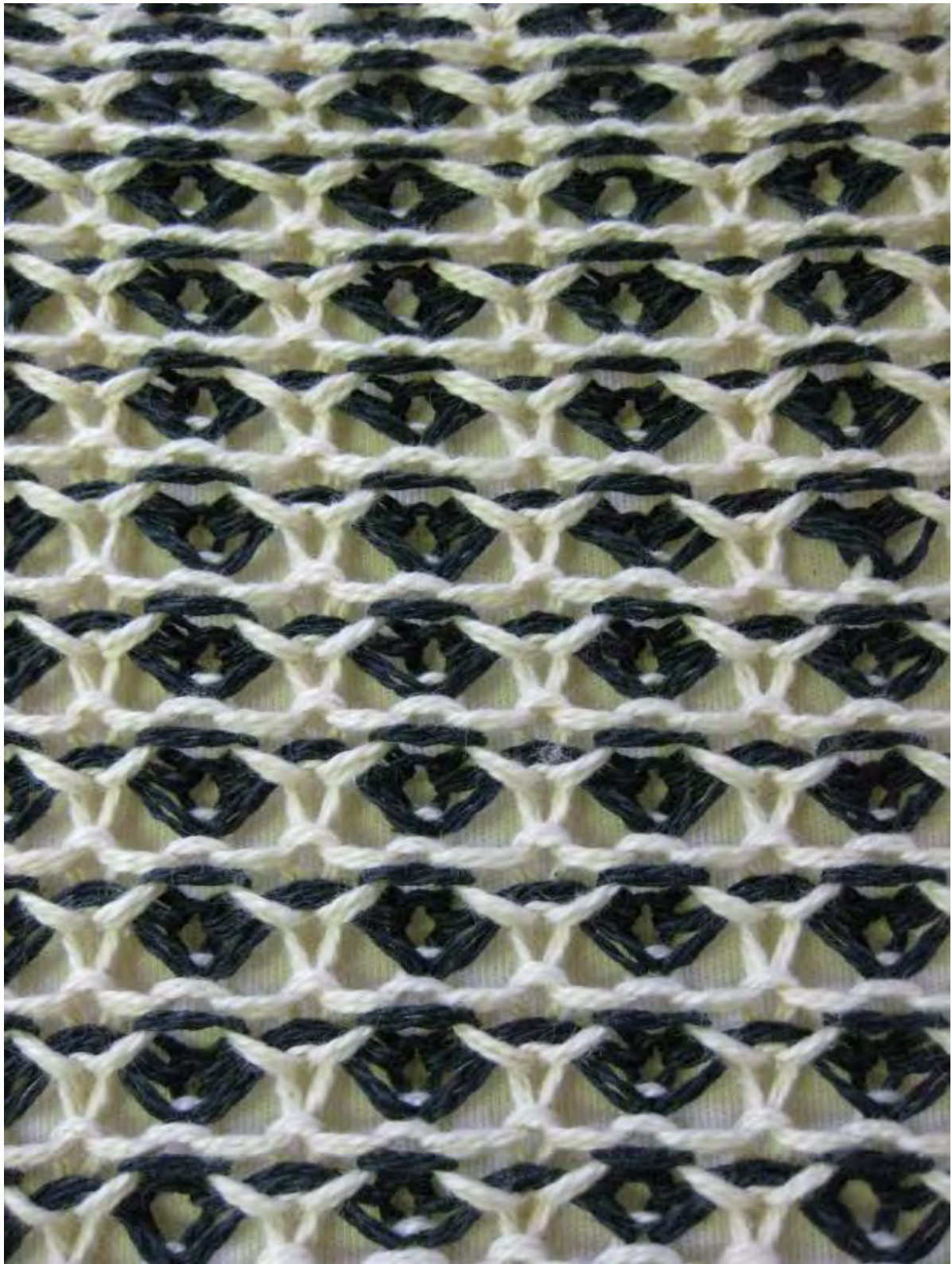
Start by hooking the side of one stitch onto one needle then closer to the centre of the sample take two, three or four sts-as in the photo- gathering the fabric.

❖ Once its all hooked on knit some more jacquard and place your new curved shape on a dress form and see where you can place it.





Hang your swatches up as you go along, you will see how your progressing!



Tuck Stitch



A tuck stitch is formed when a selected needle does not come out far enough to receive the yarn *and* make a new stitch with it, but does take the yarn into its hook on top of the existing stitch.

On a Brother machine it is the needles that come out that knit and the B position needles that tuck.

On your punchcard; punched holes knit, spaces tuck.
More than 2 spaces beside each other in a row will not work in tuck stitch.

Right side



Wrong side



- ❖ To knit; set up punchcard exactly the same as if knitting jacquard.
- ❖ Carriage on the right side Change knob to KC. Knit to the left.
Push in both Tuck buttons not MC.
- ❖ You are knitting one yarn at a time so if you want two colour as examples above you must stripe.
- ❖ So knit 2 rows yarn A, 4 rows yarn B.



- ❖ If you knit the punchcard in one colour you will get a textural sample. Experiment with combinations of stripes and maybe combining Tuck and jacquard.

❖ Notice how a tuck fabric is wider and shorter over the same number of rows as plain knit.

Slip Stitch

A slip stitch is similar to a tuck stitch, but instead of the yarn looping into the hook, it 'floats' in front of the stitch.

On a Brother machine it is the needles that come out that knit and the B position needles that slip.

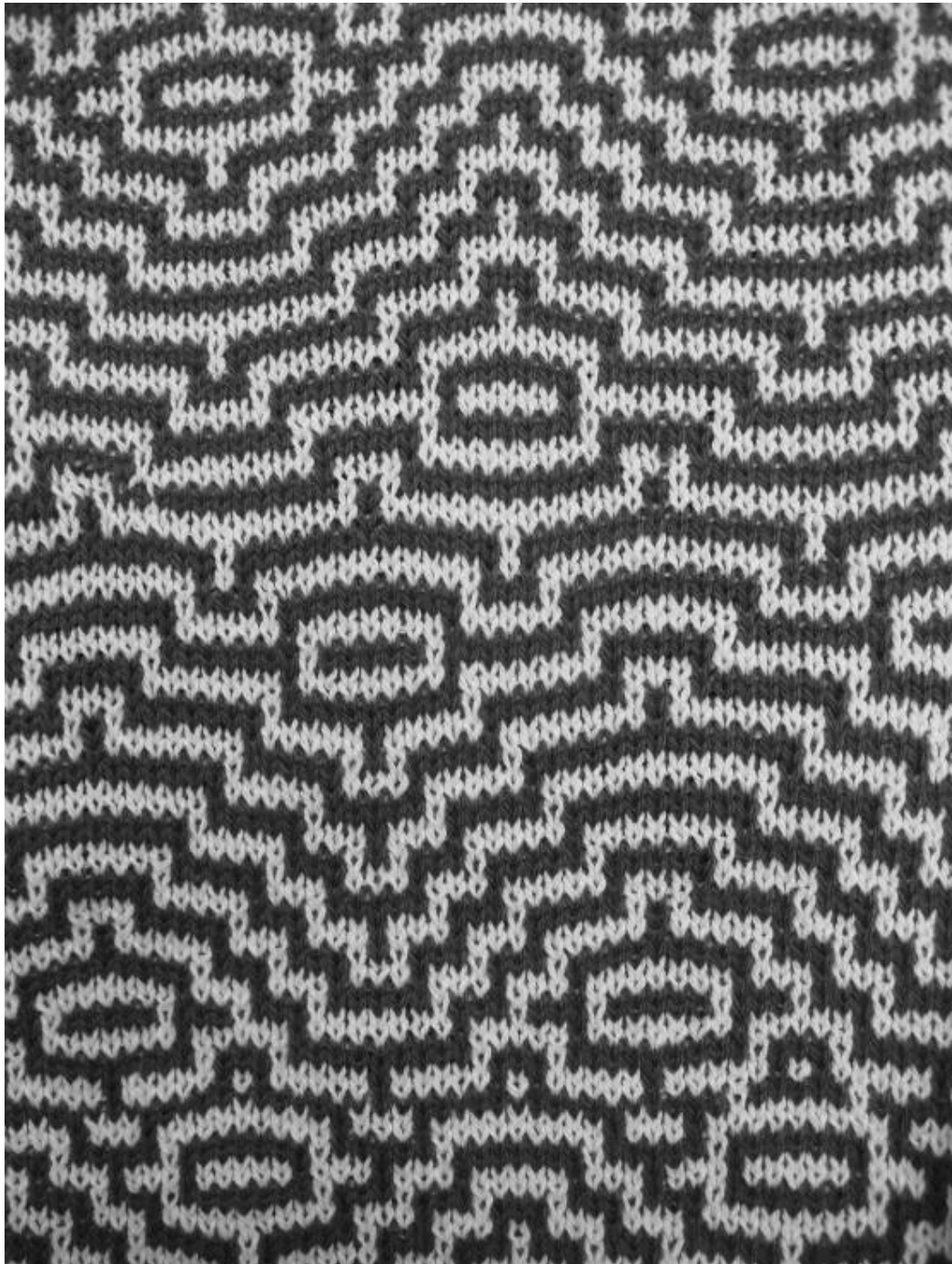


On your punchcard; punched holes knit, spaces slip.

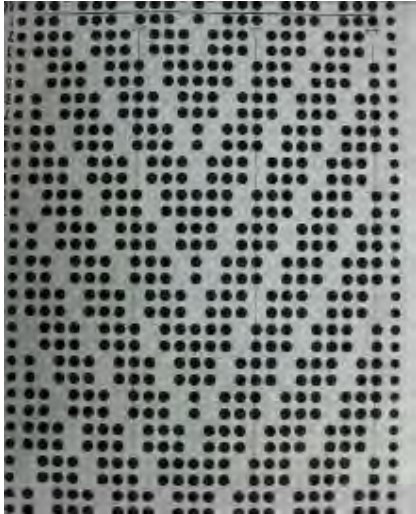
More than 4 rows slip or tuck on the same stitch may burst so put in a knit row to relieve it, or change the stitch you are slipping or tucking.

- ❖ To knit set punch card up as usual, change knob to KC to select needles.
- ❖ On the left side push in both Part buttons.

There are two Part and Tuck cams so that one can be selected to work in only the direction nearest to it – so if you only push in the left cam it will only slip/tuck when knitting to the left and vice versa. Both cams selected will slip/tuck in both directions as you have seen.



Mosaics



This is a variation of jacquard knitting –using a punchcard- incorporating either tuck or slip stitch to make a pattern.

- ❖ Creating a mosaic pattern can seem quite confusing.
- ❖ A jacquard card can be punched with a clearly visible, clearly defined geometric design, and yet the mosaic pattern will bear no resemblance whatsoever to the original pattern.

- ❖ It's obvious this card would give you a chevron when knitted as a jacquard, but knitted slip stitch in x2 row stripes it looks like the photo on the right.

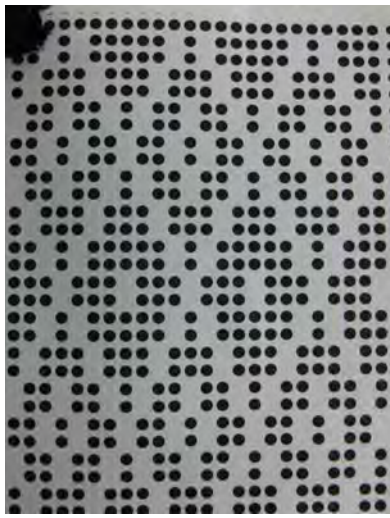


- ❖ There is no reliable way to visualise what the mosaic pattern will look like you just have to knit it.



These swatches show the right and wrong side of a mosaic fabric, showing that this technique will give you a fabric with no long floats on the reverse.





- ❖ To knit; set the card up as knitting slip stitch.
- ❖ So both part cams pushed in and knit col.A x2 rows col. B x2 rows.
- ❖ Knit through the card striping like this.

In some patterns certain needles must slip or tuck for several successive pattern rows, resulting in long vertical stitches and at the same time forcing some of the horizontal rows to have a diagonal or curved appearance.



- ❖ It is important that no needle tuck or slip more than six or eight rows, or the fabric will become too unbalanced and will be difficult to knit.



- ❖ Here is a punchcard and sample done on both elongate and on slip stitch, the card looks very different to the knitted samples!



Make a moodboard of trends that go together with your samples!



Engineering



You can place your pattern wherever you want on the body and also on your fabric.

- ❖ To knit; after your KC row and your needles are selected.



- ❖ Push back needles wherever you don't want the pattern.

- ❖ Here I have done it randomly so it gives a look as if the jacquard has been 'rubbed out'.



- ❖ The long floats on the back could also be used as part of a design.

You can also do this in a controlled way to give you plain and patterned blocks.

- ❖ Push the same block of needles back for the same number of rows as are needles.



Fabric Manipulation

- ❖ You can further develop your fabric by washing.
 - ❖ This lambswool sample was distorted.
 - ❖ Tie with thick synthetic yarn in some places not others –as you would for tie-dyeing.
 - ❖ Machine wash.
 - ❖ This will cause the tied areas not to change and the rest to shrink and felt.
-
- ❖ Try engineered ladders.



- ❖ Push back the needles that would have formed the centre of the diamond.
- ❖ Transfer the stitches and push the needles back out of action so they will ladder.
- ❖ Bring them back in again as the diamond closes.

- ❖ You could do this in a controlled way like this or more randomly.



- ❖ Use ladders and the jacquard together to create a very open look.



- ❖ Try using traditional patterns such as dog-tooth and engineer the pattern for a modern look. Black and white looks great and also using thick and thin yarns together and washing them creates a textured effect.





- ❖ Cutting the floats on a jacquard once washed looks great and creates a “fur-like” texture.



Single Colour Jacquard



- ❖ If you want a more organic, textural look try knitting a jacquard in two yarns of the same colour but different qualities.
- ❖ This example is lambswool and cotton. So if you machine wash the sample the wool will felt/shrink and the cotton will not.
- ❖ Or as I mentioned before use a tuck stitch in one colour.

- ❖ Try a thick yarn and a thin yarn, cut the floats and use as part of your design.
- ❖ Try using one colour or 2 different colours to create texture and a fur look.



So experiment with your punchcards and the possibilities are endless!

Glossary

E-wrap *cast on method for finished edge loop yarn around needles from left to right making the shape of a lower case e around each needle while needles are in E position.*

Sts *stitches(abbreviation)*

Prong tool *double ended tools that come with machine either have 1 and 2, 1 and 3, 2 and 3 prongs.*

Latch tool *tool that looks like a machine needle, used for picking up stitches.*

On the left side of the needle bed on machine

Needle position;

A *non working position/ needles will not knit*

B *working position / needles will knit*

D *selected for pattern knitting/ needles will knit*

E *holding position/ needles will knit unless holding cam is switched to H or II*

5 gauge *5 needles to the inch*

HP *holding position*

Holding cam *control on carriage on Brother machines N
Or H on Knitmaster I or II. Select H and II for short row knitting, needles selected into position E will NOT knit.*

Swatch *piece of knitting/fabric*

Pointelle *lace knitting*

Eyelet *one stitch lace hole*

1x1 pointelle *1 row of 1 stitch lace holes*

Fashion / fashioning *moving multiple stitches*

Cast off *bind off*

Dress Form *dummy/ mannequin*



Here is our story , well a particular part of it

Sue and I met at a jumble (that`s second hand clothes) sale in our village, Old Basing near Basingstoke, Hampshire in 1978 !!!

After our respective University and Polytechnic Courses we set up in Brighton in 1983. We called our label "Pretty Blue Guns" after one of our favourite Tom Waits` tracks. We were lesser known contemporaries of Body Map, John Flett, Martin Kidman of Joseph and all that crowd of British design hopefuls eons back at the dawn of time.

We were marvellously supported and encouraged by Steve & Iona Hudson, of New Masters, Kings Road, London and then by Stuart Malloy & Peter Sidell (now owner of The Library, in South Kensington) of Jones of Kings Road and later Floral Street. We had US sales to Macys and Neiman Marcus and various other accounts but after about 3 years we couldn`t really make the line pay, nevertheless our garments had come to the notice of an Italian company called Agenco of Modena and as a result we got great experience for the first time working in Italy on one of their knitwear lines called Portobello`s. This and Sue`s earlier employment in the Far East with US brand Williwear gave us the necessary commercial understanding to separately work for Next , Henry Lloyd and Boules. All these were very useful experiences.

Pretty Blue Guns had also been noticed by our now long-time friends Chris Matthews & Philippa George of` The Collection`, the design studio for whom we started producing one off knit designs in 1987 and still work with today. Our own design collection started with a first trip to New York in 1991 under the catchy banner of our combined surnames `Dykes Enticknap` which luckily in year 2000 with the on-set of the dot com age was transmuted into the rather more generic user friendly Knit-1 Ltd www.knit-1.com . Design sales were growing rapidly and we needed more space so moved to our existing premises, a 3,500 sq ft former engineering works which we completely refurbished to our design business requirements. We have sold designs to and worked with most of the big global fashion brands.

In 2008 we set up our Design Courses www.knitdesigncourses.com . Somewhat frustrated by the practical skills levels which were lacking in the portfolios of knit graduates and undergraduates we had been seeing for some time we set about devising our intensive courses. We wanted to offer a completely new way to learn knit design & make that would appeal to not just students, technicians and tutors but industry professionals, ambitious designers and knit design enthusiasts. In fact the courses appeal to any aspiring individual wanting to progress in the field of machine knit design and knitwear for the fashion industry.

The courses take place in the design business environment offered by Knit-1 Ltd at the company's well equipped premises in Brighton, UK. Many elements of the courses have references to selected high profile customers of the company which further distinguishes the courses from what can be offered by traditional academic institutions. Hence we can deliver the required practical skills to advanced levels in knitwear, knit design & make, print and embroidery in rapid time.

The courses are now very well established and have proved themselves through the positive feedback received and the attainment of jobs in the fashion industry by course participants. It takes the knowledge of a fashion business to know what other businesses want from their prospective employees. Furthermore many of our participants have also set up their own businesses after taking one of our courses.

That's it so far.....

www.knit-1.com

www.knitdesigncourses.com

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